Visual Studio LIVE!
EXPERT SOLUTIONS FOR ENTERPRISE DEVELOPERS | Las Vegas

# Zero to Blazor Apps with GitHub Copilot


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Level: Introductory to Intermediate

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# Session Survey

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- Use the QR code or search for “Converge360 Events” in your app store
- Find this session on the Agenda tab
- Click “Session Evaluation”
- Thank you!

# Agenda

- GitHub Copilot
  - Core features and value
  - Get up and running
  - Chat/Agent Mode
  - Understanding code context
- AI, LLMs and Prompt Engineering
- Zero to a Web App for Blazor
  - Leverage GitHub Copilot to build an app!

# Welcome to GitHub Copilot

- AI-powered code assistant
- Real-time coding assistance
- Contextually aware
- Multiple IDE integrations
- Multi-Language Support
- Powered by mainstream LLM models
- Next Evolution in Software Engineering

# GitHub Copilot | Features

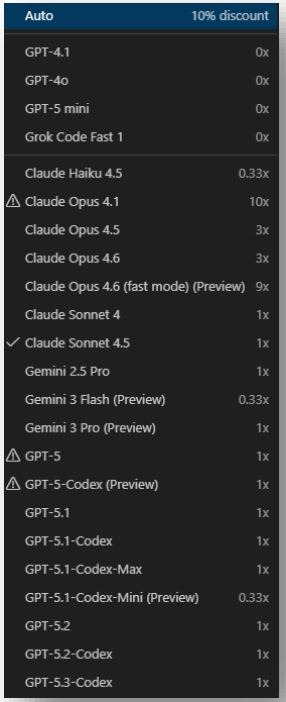
- Code generation
- Inline code assistance
- Immersive chat
- Code explanations
- Unit testing
- Code refactoring
- Code optimizations
- Fix broken code
- Legacy migration
- Document code
- Secure coding
- Code translation
- Pull request summaries
- Code reviews

# GitHub Copilot | Value

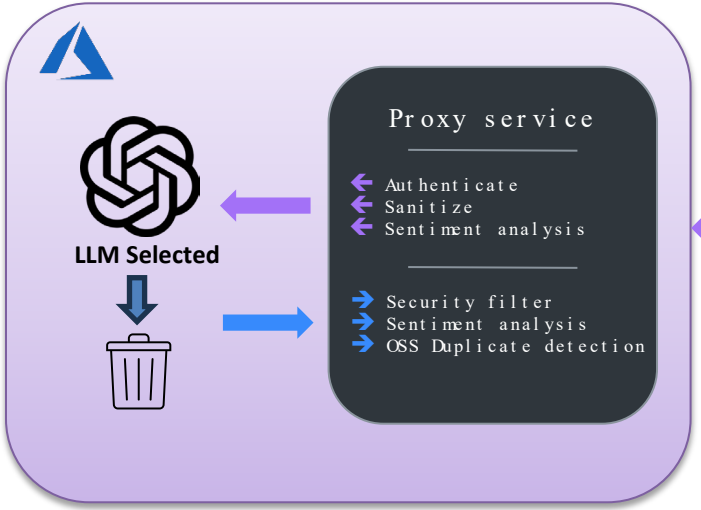
- Increased Productivity
- Accelerate Coding Speed
- Stay in the Flow
- Developer Experience (DevEx)
- Amplify Potential
- Compress the Learning Curve
- Reduce Onboarding Time
- Improves Code Quality
- Saves Time and Frustration

# GitHub Copilot | AI & LLMs

- OpenAI (GPT), Anthropic (Claude), Google (Gemini) Models
  - Ability to toggle models*
  - Newer models adopted by Copilot
  - Tailored for technical prompts only
- Probabilistic not Deterministic
- Transformer architecture

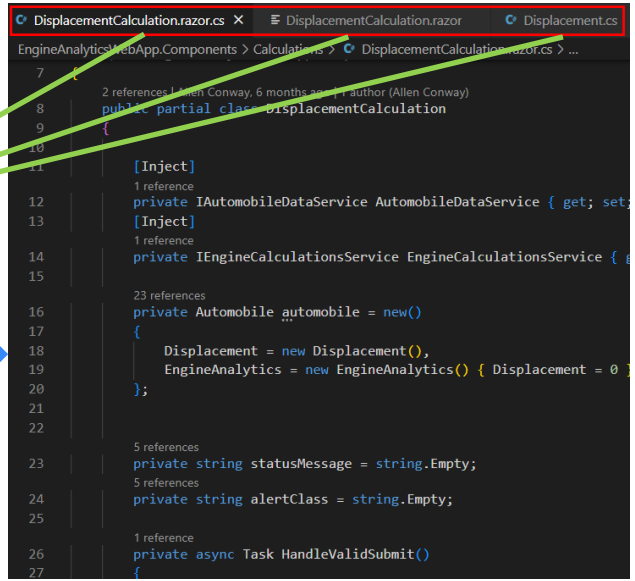


# GitHub Copilot | How it Works




Context / Prompt

Suggestions

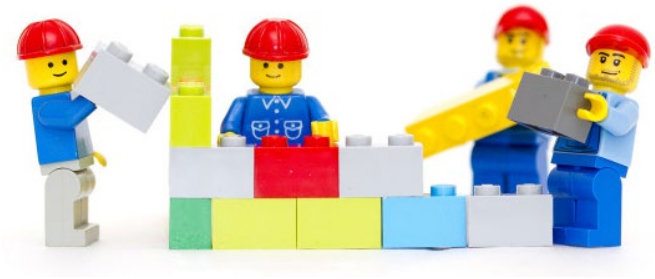


# GitHub Copilot | Getting Started

- Copilot License + Account + IDE
- Supported IDEs
  - Visual Studio Code
  - Visual Studio
  - Jet Brains
  - Vim/Neovim
  - Xcode
- Install extensions / plugin
- PSA: Keep IDE and extensions up to date!

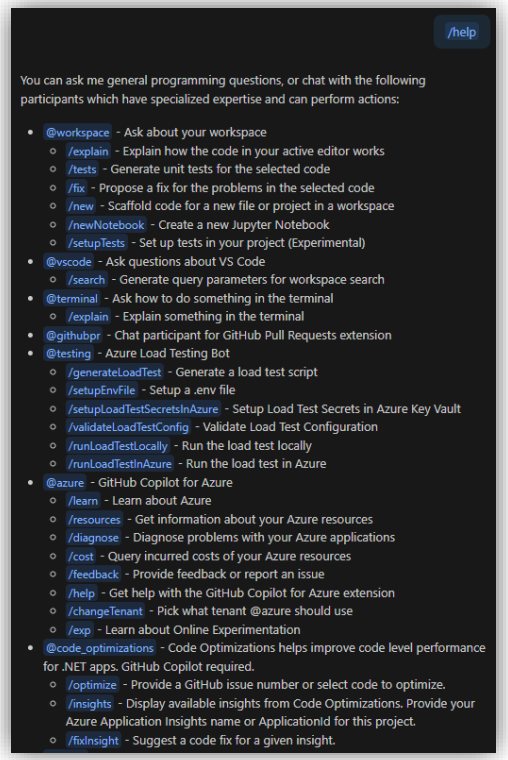
# GitHub Copilot | New Blazor App

- Leverage Chat for creating a new Blazor .NET 10 app
- Methods for scaffolding
  - Prompt Chat for instructions on templates
  - Use `/new` to scaffold an app
  - Agent mode from scratch
  - Use IDE with templates built in



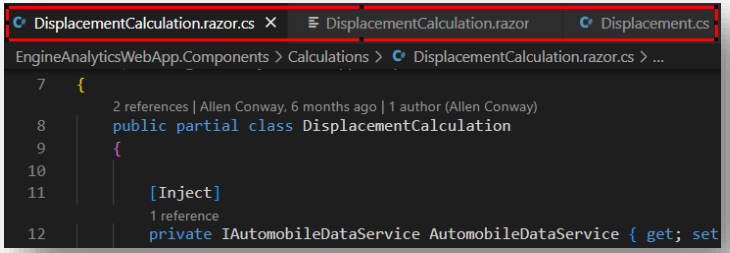
# GitHub Copilot Ask | Agents & Slash Commands

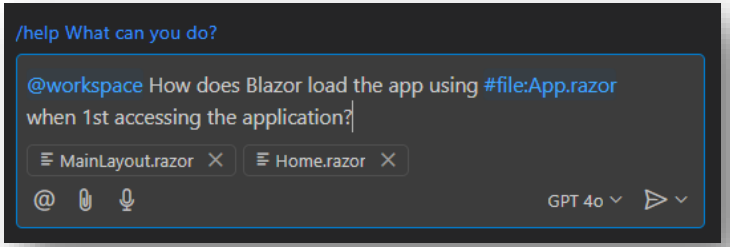
- Custom **agents** provide targeted assistance
- **Slash** commands interact with agents that provide shortcuts to perform routine tasks or generate code
- Use `/help` to see commands for your IDE



# GitHub Copilot | Code Context

- Mix of explicit and dynamic discover of code
  - Depends on Chat mode
- Aim for relevant code
  - Inline / Ask mode
- Providing context *when necessary* improves results



# Prompt Engineering

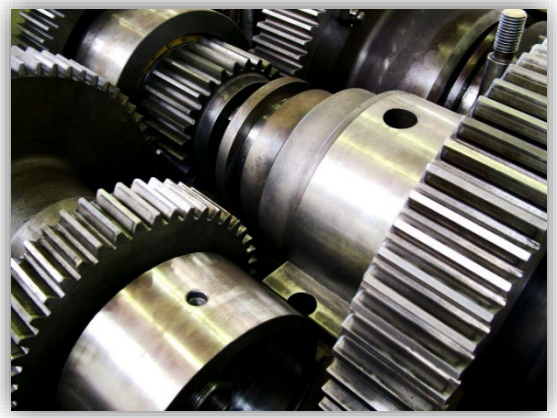
- Crafting of user input to guide AI to desired outputs
- Yields most accurate and desired results
- Ability to provide in-context learning
- Reduces iteration
- Allows getting the most out of LLMs today

# GitHub Copilot | Add Functionality

- Copilot offers capability to scaffold new code and features
  - **Ask** / **Plan** for clarity
- Copilot **Agent Mode**
  - Multi-step coding tasks
  - Autonomous/Semi-Autonomous

# GitHub Copilot | Code Refactoring

- Leverage Copilot to refactor code
  - High cyclomatic complexity
  - Monolithic code
  - Improve performance
  - Security audit
  - Apply known patterns
  - Legacy code



# GitHub Copilot | Unit Testing

- A crown jewel of AI and Copilot
  - Saves time especially with complex code
- Ability to generate new tests
  - `/tests` | Agent mode
- Add in additional details
  - Mocking
  - Positive/negative scenarios

# GitHub Copilot | Fixing Code

- Copilot is available in all aspects of development including troubleshooting
  - Fixing code
  - Debugging errors
  - Analyzing build issues
- Copilot Agent mode
  - Provide error output
    - Browser, logs, terminal, build server, etc.

# GitHub Copilot | Infrastructure as Code

- Generate Terraform files
  - Terraform MCP server
- Generate .yaml build pipeline for GitHub Actions
- Leverage Copilot Agent mode
  - Code generation
  - Security questions

# Useful Links

**GitHub Copilot Features**

https://docs.github.com/en/copilot/about-github-copilot/github-copilot-features

**GitHub Copilot Documentation**

https://docs.github.com/en/copilot

**Configuring GitHub Copilot (per IDE)**

https://docs.github.com/en/copilot/managing-copilot/configure-personal-settings/configuring-github-copilot-in-your-environment

**GitHub Next**

https://githubnext.com/

**Take GitHub Copilot on a test-flight**

https://resources.github.com/copilot-demo/

# Thank you! Q&A

http://c1.staticflickr.com/1/28/65098350_b7bd96f38_b.jpg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